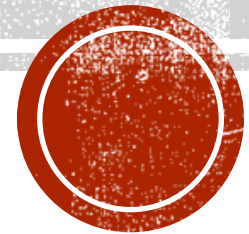


COMBINATIONAL CIRCUITS/

Digital Electronics



8 X 1 MUX

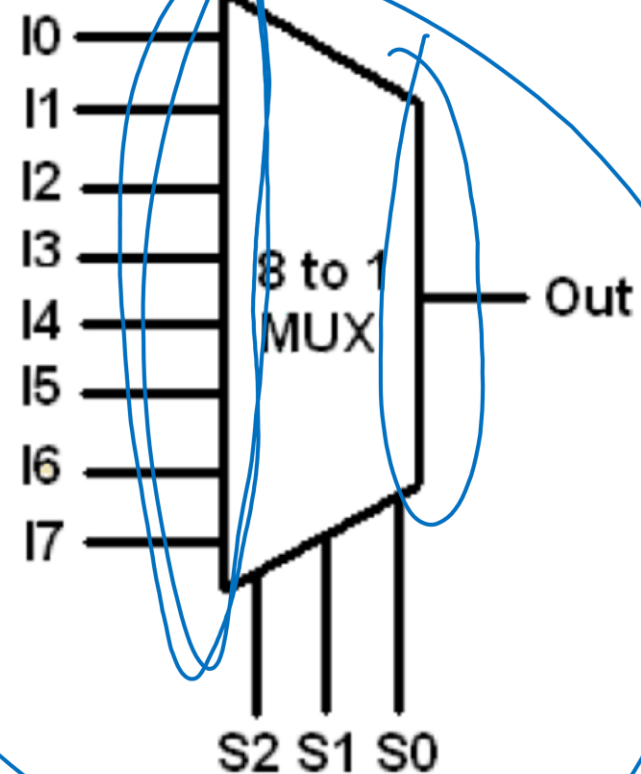
Example 1

A7
A6
A5
A4
A3
A2
A1
A0

8x1
Multiplexer

Y

S₁ S₀ S₀
Select lines



WORKING

- An 8:1 multiplexer (MUX) is a combination logic circuit which has 8 input lines ($I_0, I_1, I_2, I_3, I_4, I_5, I_6$, and I_7).
- 3 select line (S_0, S_1 , and S_2), and 1 output line (Y).
- The logic levels applied to the select lines S_0, S_1 , and S_2 determine which data input will transmit to the output line.
- To determine the logic expression for output (Y) of the 8:1 multiplexer and its logic circuitry,
- we require its function table (truth table) which provides information about the operation of its logic circuit.

TRUTH TABLE

Select Lines			Output (Y)
S_2	S_1	S_0	
0	0	0	A_0
0	0	1	A_1
0	1	0	A_2
0	1	1	A_3
1	0	0	A_4
1	0	1	A_5
1	1	0	A_6
1	1	1	A_7

m_0
 m_1
 m_2
 m_3
 m_4
 m_5
 m_6
 m_7

FULL TRUTH TABLE (ACTUAL)

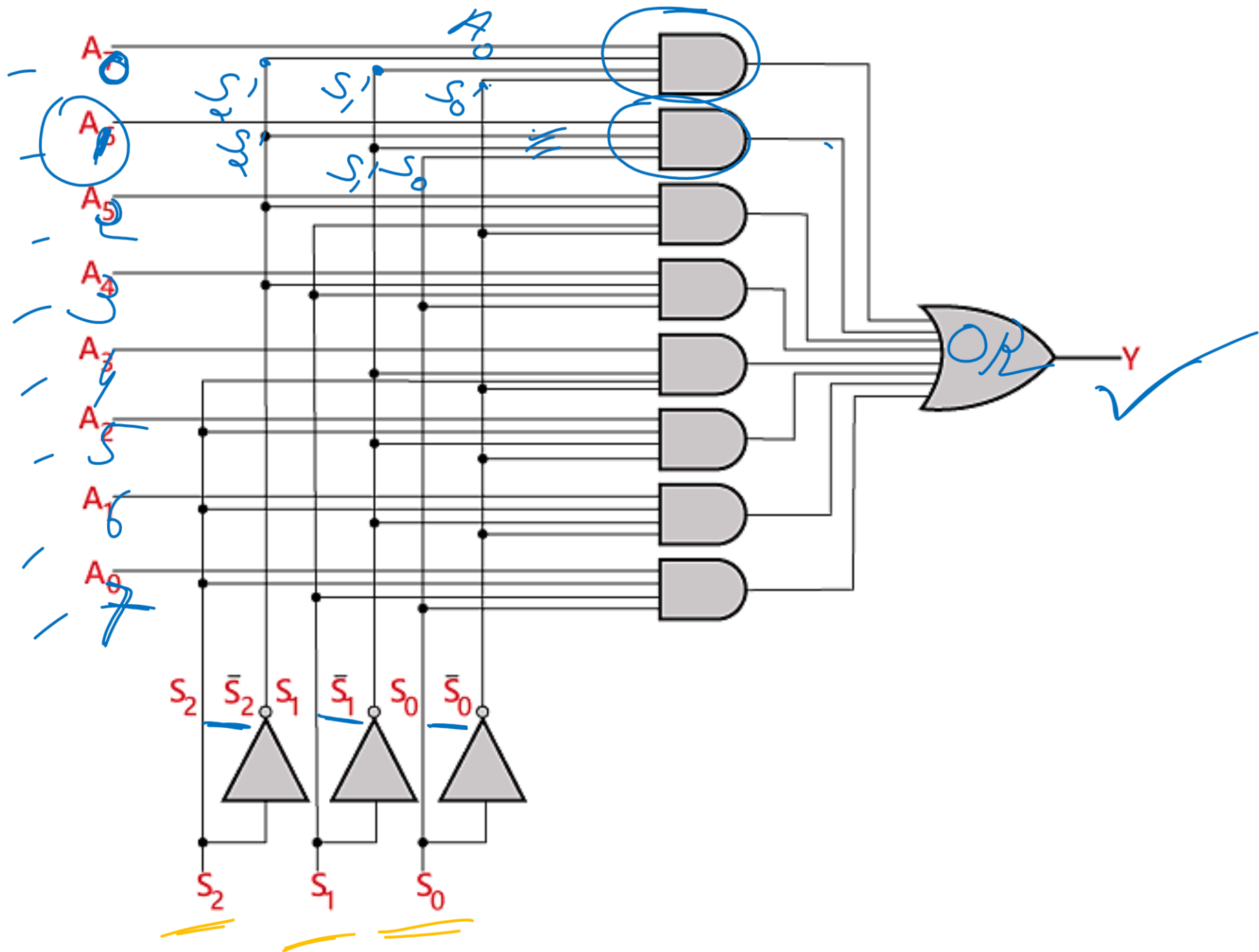
Select			Inputs								Output
S2	S1	S0	H	G	F	E	D	C	B	A	
0	0	0	X	X	X	X	X	X	X	1	1
0	0	1	X	X	X	X	X	X	1	X	1
0	1	0	X	X	X	X	X	1	X	X	1
0	1	1	X	X	X	X	1	X	X	X	1
1	0	0	X	X	X	1	X	X	X	X	1
1	0	1	X	X	1	X	X	X	X	X	1
1	1	0	X	1	X	X	X	X	X	X	1
1	1	1	1	X	X	X	X	X	X	X	1

SOP $\Rightarrow$ 1
 $\Sigma = m_0 + m_1 + m_2 + \dots + m_7$

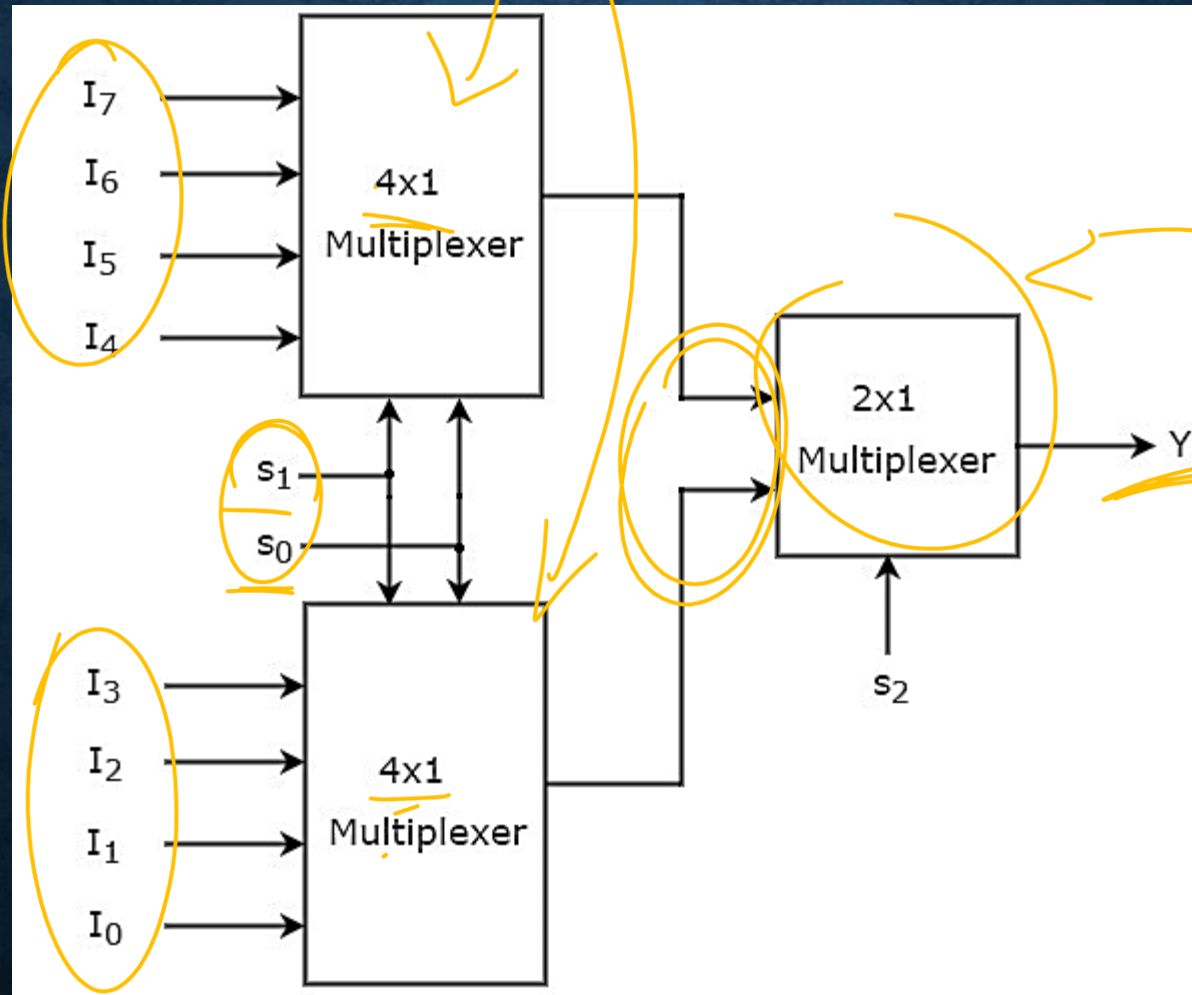
m_0
 m_1
 m_2
 m_3
 m_4
 m_5
 m_6
 m_7

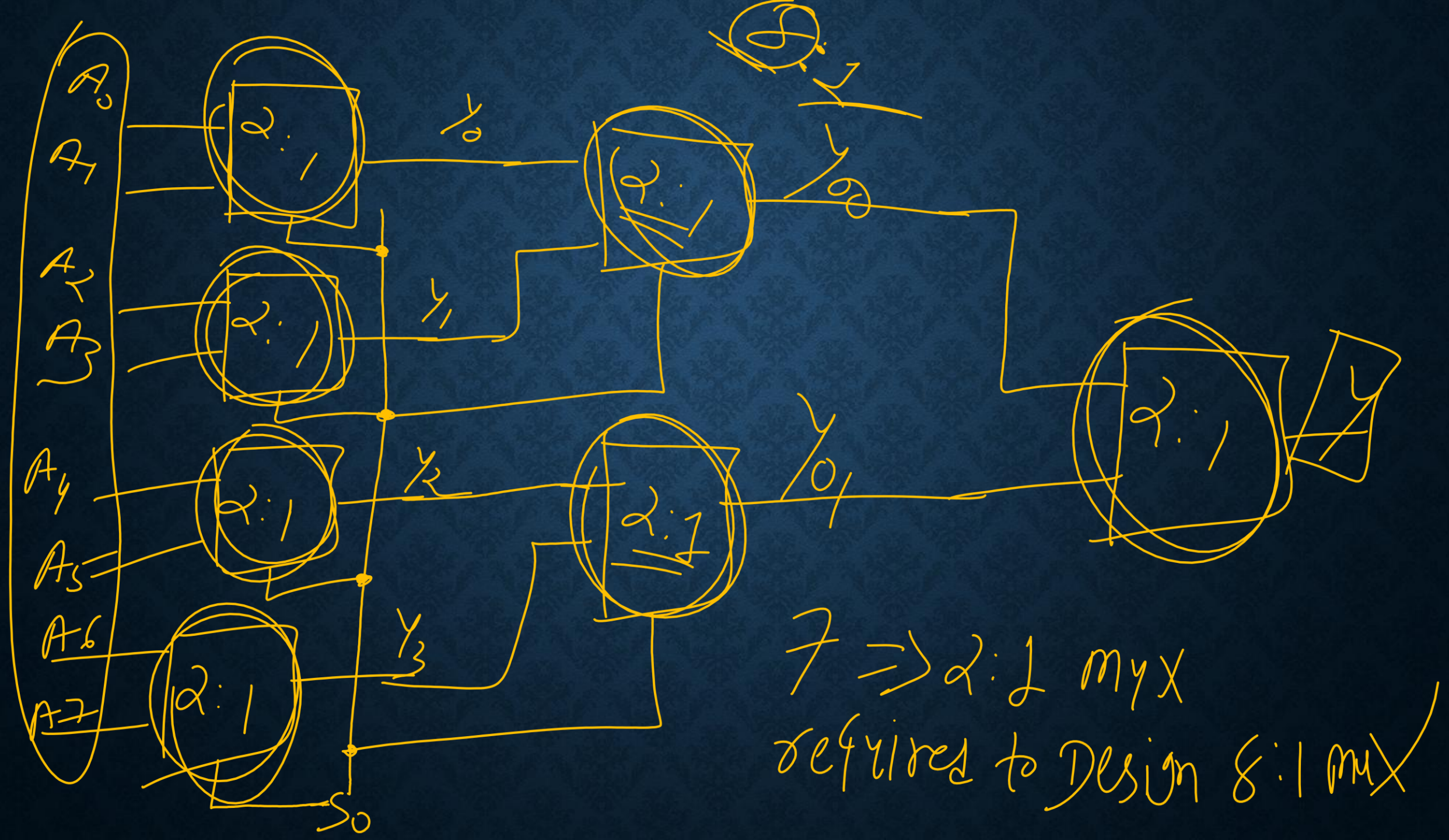
$$y = m_0 + m_1 + \dots + m_7$$

$$\begin{aligned}
 Y = & \underbrace{S_0' \cdot S_1' \cdot S_2'}_{\checkmark} \underbrace{A_0}_{\checkmark} + \underbrace{S_0 \cdot S_1' \cdot S_2'}_{\checkmark} \underbrace{A_1}_{\checkmark} + \underbrace{S_0' \cdot S_1 \cdot S_2'}_{\checkmark} \underbrace{A_2}_{\checkmark} + \\
 & \underbrace{S_0 \cdot S_1 \cdot S_2'}_{\checkmark} \underbrace{A_3}_{\checkmark} + \underbrace{S_0' \cdot S_1' \cdot S_2}_{\checkmark} \underbrace{A_4}_{\checkmark} + \underbrace{S_0 \cdot S_1' \cdot S_2}_{\checkmark} \underbrace{A_5}_{\checkmark} + \\
 & \underbrace{S_0' \cdot S_1 \cdot S_2}_{\checkmark} \underbrace{A_6}_{\checkmark} + \underbrace{S_0 \cdot S_1 \cdot S_3}_{\checkmark} A_7
 \end{aligned}$$



8 X 1 MUX USING 4 X 1 MUX

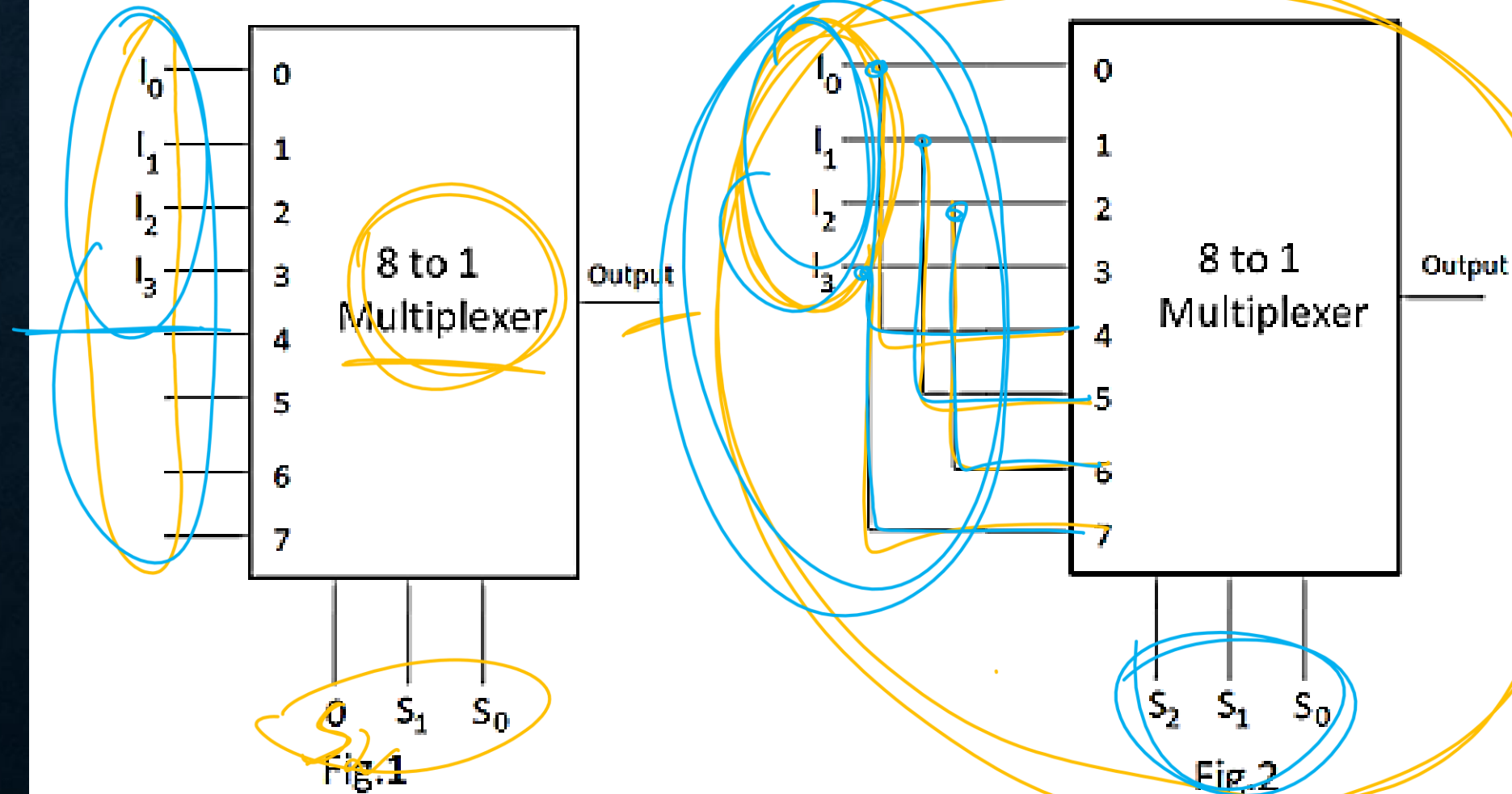




CONVERSION OF 8 X 1 MUX TO 4 X 1 MUX

Short I/P Lines to half

Conversion of 8 to 1 Multiplexer to 4 to 1 Multiplexer

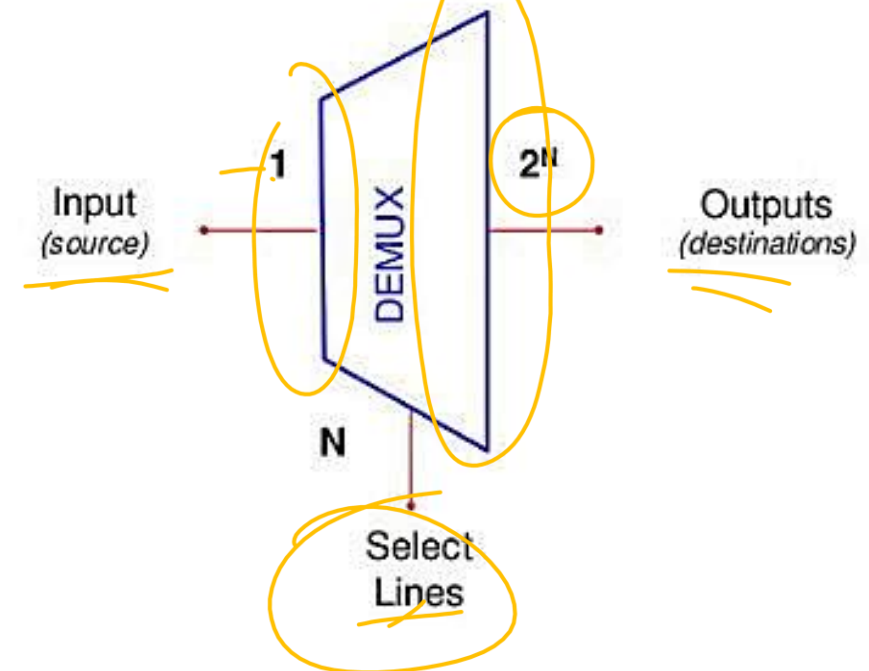
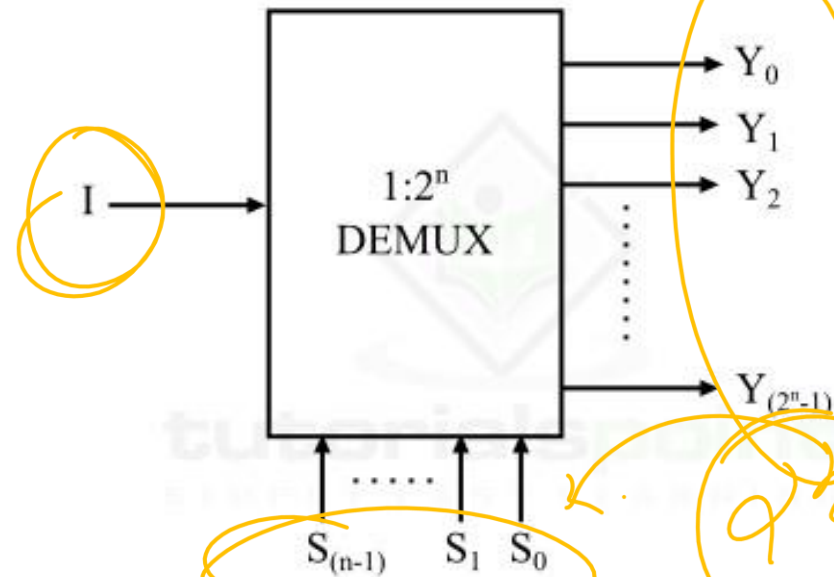


DEMUX

- A Demultiplexer is a combinational logic circuit that accepts a single input and distributes it over several output lines.
- As Demultiplexer is used to transmit the same data to different destinations, hence it is also known as data distributor.
- Demultiplexer has only one data input line, 2^n output lines, and n select lines.

Part B $\Rightarrow$ (14/15 Lectures till today)

BLOCK DIAGRAM



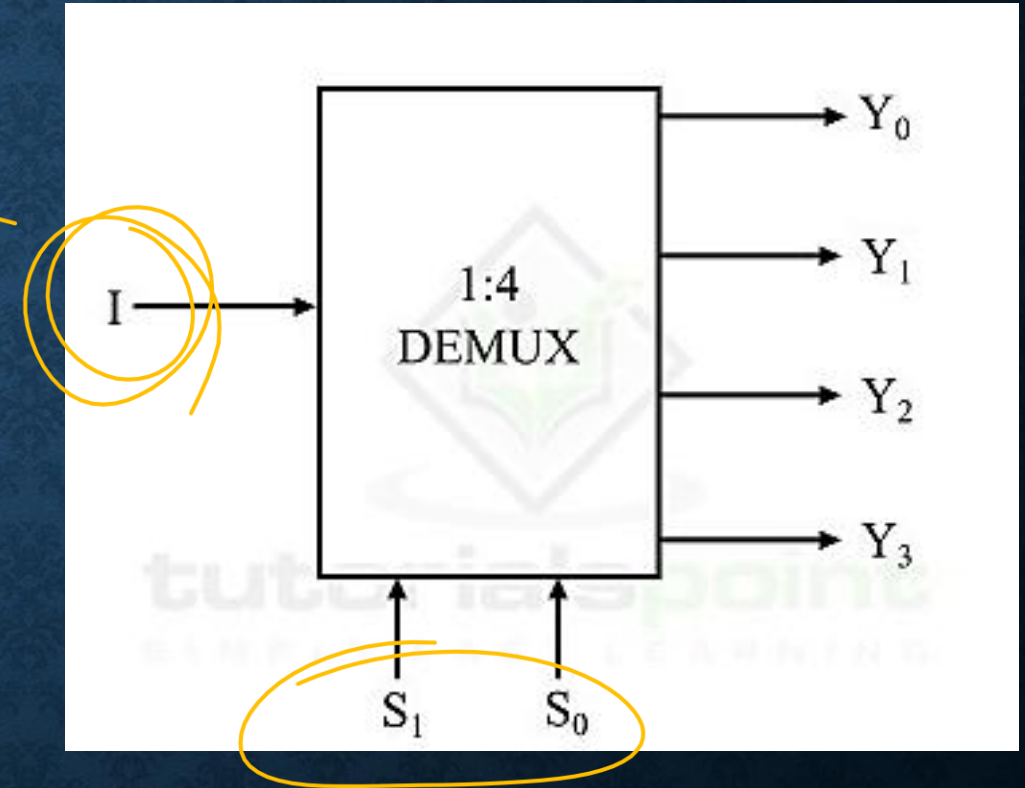
Content $\rightarrow$ Live videos $\Rightarrow$ Wg Ysg sg.

TYPES OF DEMULTIPLEXERS:

- 1-to-2,
- 1-to-4,
- 1-to-8,
- 1-to-16, and
- 1-to-32, based on inputs

1×4 DEMULTIPLEXER

- The 1×4 Demultiplexer has 1 input line (I), 2 select line (S_0 and S_1), and 4 output lines (Y_0 , Y_1 , Y_2 , and Y_3).
- The logic level applied to the select lines determines the output line to which the input data (I) will be transmitted.



TRUTH TABLE

Select Line		Outputs			
S_1	S_0	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0

• $Y_0 = S_1^- S_0^- I$

• $Y_1 = S_1^- S_0 I$

• $Y_2 = S_1 S_0^- I$

• $Y_3 = S_1 S_0 I$

INTRODUCTION TO PARITY BITS

Error Detection: The Parity Bit

- **Definition:** A redundant bit added to a binary word to check for errors during transmission. *Signal FFFF*
- **The Problem:** Electrical noise or interference can flip bits (0 to 1 or 1 to 0).
- **The Goal:** To ensure the data received at the destination matches the data sent from the source.
- **Key Benefit:** Simple, low-cost, and widely used in digital systems.

TYPES OF PARITY

Even vs. Odd Parity

- **Even Parity:**

- The parity bit is set so that the total count of 1s (including the parity bit) is **Even**.

- **Odd Parity:**

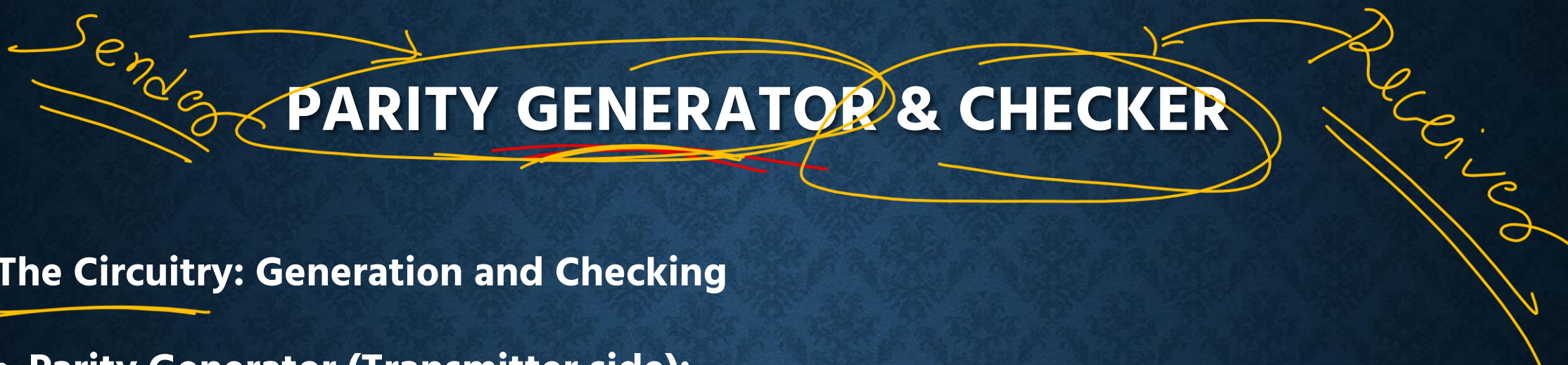
- The parity bit is set so that the total count of 1s is **Odd**.

Data (7-bit)	<u>Number of 1s</u>	<u>Even Parity Bit</u>	Odd Parity Bit
1011001	4 (Even)	0	1
1010100	3 (Odd)	1	0

ODD & EVEN PARITY BITS FOR DECIMAL DIGITS FROM 0 TO 9

Decimal Digit	BCD	Even Parity	Odd Parity
0	0 0 0 0	0	1
1	0 0 0 1	1	0
2	0 0 1 0	1	0
3	0 0 1 1	0	1
4	0 1 0 0	1	0
5	0 1 0 1	0	1
6	0 1 1 0	0	1
7	0 1 1 1	1	0
8	1 0 0 0	1	0
9	1 0 0 1	0	1

PARITY GENERATOR & CHECKER



The Circuitry: Generation and Checking

- Parity Generator (Transmitter side):

- A combinational logic circuit that calculates the parity bit based on the data.

- Parity Checker (Receiver side):

- Counts the 1s in the received message.
- If the result doesn't match the agreed-upon parity, it signals a **Transmission Error**.

IMPLEMENTATION USING LOGIC GATES

Logic Gates: The Role of XOR

- **The Logic:** The sum of an **odd** number of 1 is always 1; the sum of an **even** number of 1 is always 0.
- **XOR Property:** An Exclusive-OR gate outputs a 0 when there is an even number of 1s at the input.
- **Hardware:** Parity circuits are built by cascading XOR gates together.

$\Rightarrow 2+2=4$ I/P $0+1=1$ O

$\rightarrow$ min 2 I/P

EVEN PARITY GENERATOR (3-BIT MESSAGE)

An even parity generator ensures that the total number of 1s in the transmitted block (Data + Parity Bit) is an **even number**.

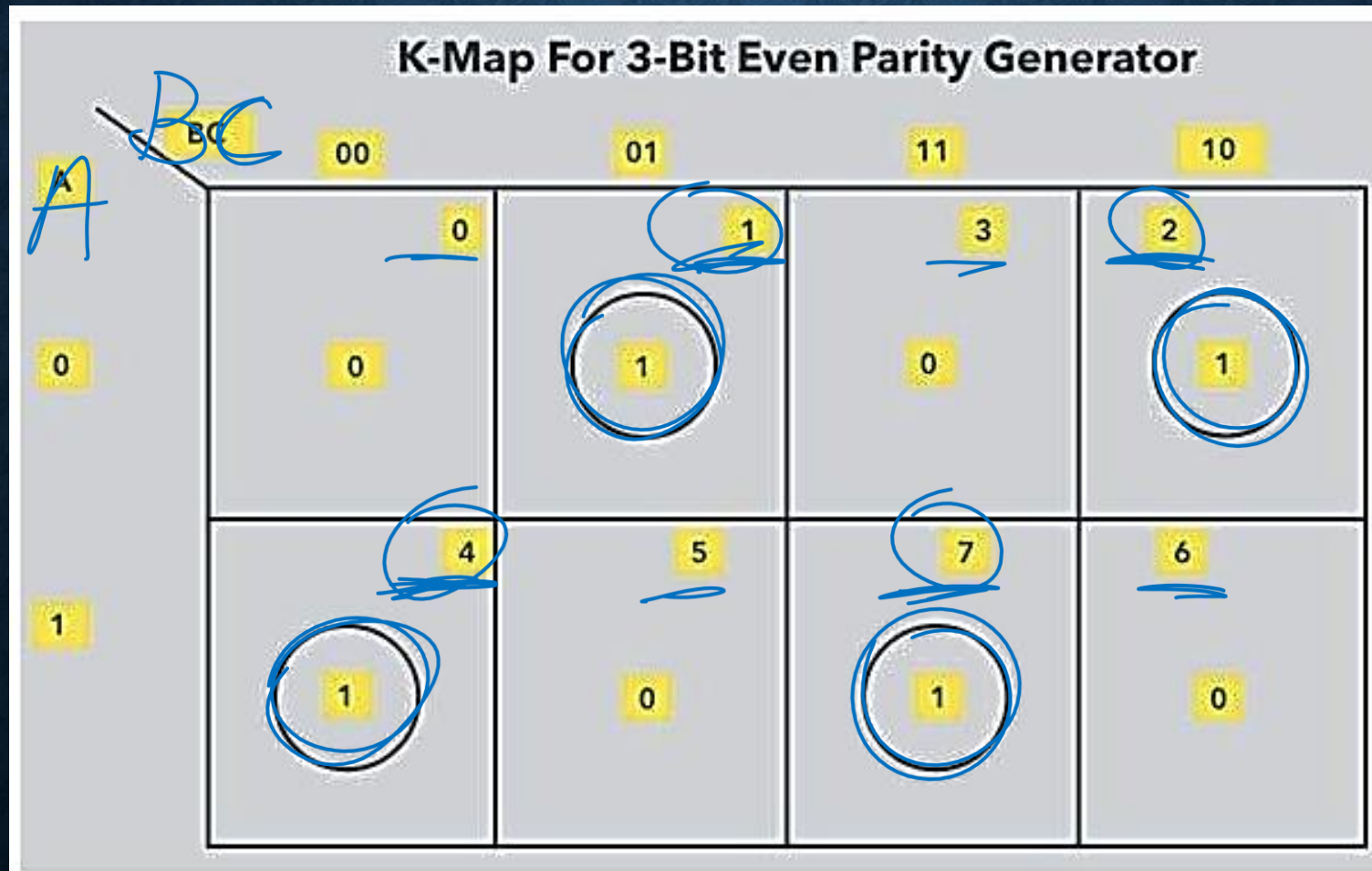
TRUTH TABLE

For a 3-bit input (A, B, C), the parity bit (P) is generated as follows:

Input A	Input B	Input C	Even Parity Bit (P)	Total 1s (incl. P)
0	0	0	0	0 (Even)
0	0	1	1	2 (Even)
0	1	0	1	2 (Even)
0	1	1	0	2 (Even)
1	0	0	1	2 (Even)
1	0	1	0	2 (Even)
1	1	0	0	2 (Even)
1	1	1	1	4 (Even)

~~Full Adder~~
 ~~$A \oplus B \oplus C$~~
 $\Rightarrow A'B'C' + A'B'C + AB'C' + ABC$

K - MAP SIMPLIFICATION



LOGIC IMPLEMENTATION

- From the truth table, we can see that the parity bit P is high (1) only when there is an odd number of 1s in the input.
- This is the exact definition of the Exclusive-OR (XOR) operation.
- The Boolean expression for a 3-bit even parity generator is:

$$P = A \oplus B \oplus C$$

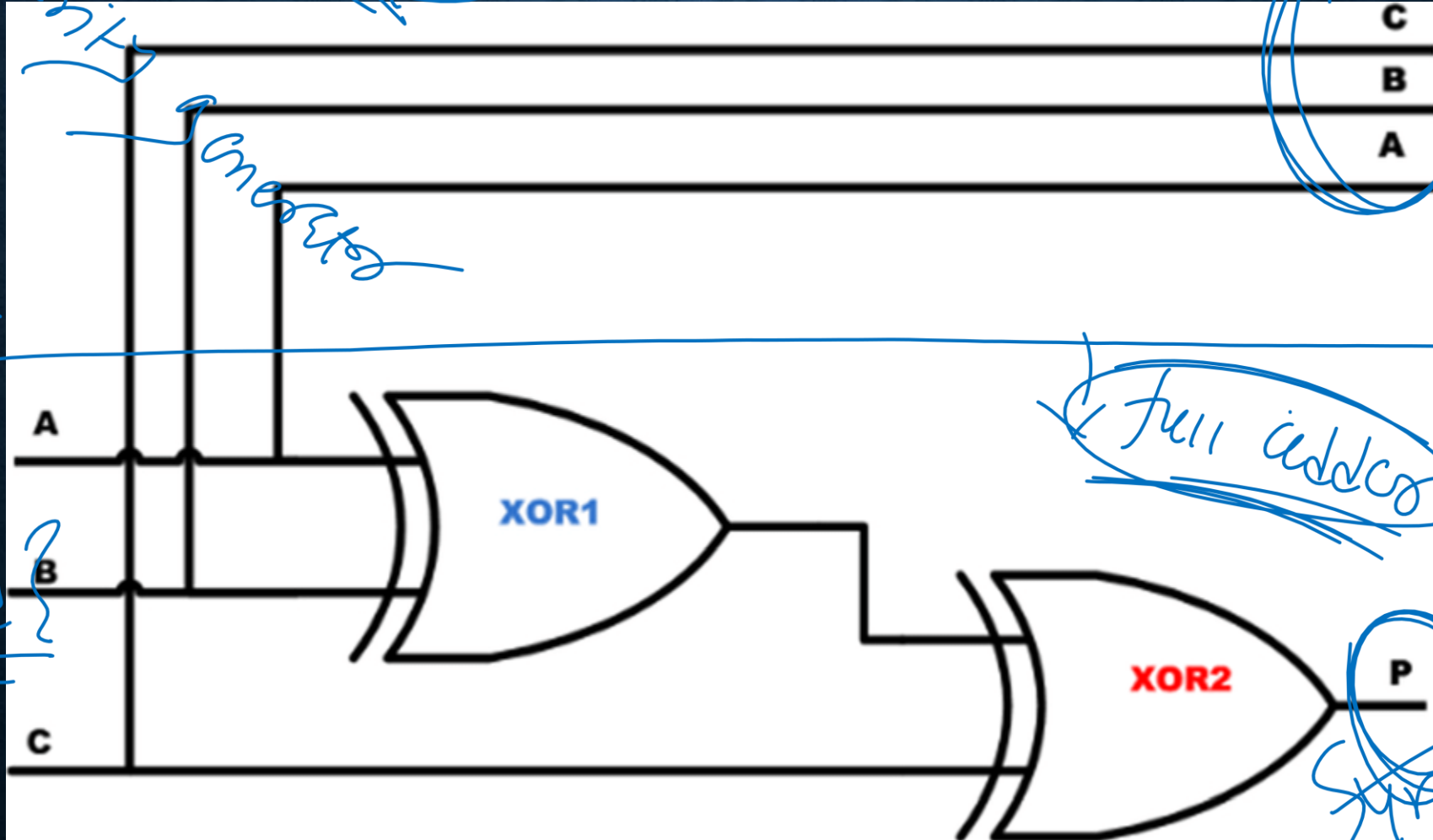
full adder circuit

4 bit IP $\Rightarrow$ even parity expression = (?)

$$A \oplus B \oplus C \oplus D \oplus E$$

MCQ

CIRCUIT DIAGRAM



EVEN PARITY CHECKING (RECEIVER)

- **Definition:** A digital error-detection method that ensures the total number of **1s** in a binary string (data bits + parity bit) is **even**.
- **The Parity Bit:** An extra bit added to the message at either the **MSB** (Most Significant Bit) or **LSB** (Least Significant Bit) position.
- **The Checker's Goal:**
 - Receive the binary code.
 - Count the number of **1s**.
 - If the count is **even**, the code is "Error-Free."
 - If the count is **odd**, an error has occurred during transmission.



CASE STUDY 1 – ERROR-FREE TRANSMISSION

- **Scenario:** A 4-bit message with an LSB parity bit.
- **Received Code:** 00110
- **Breakdown:**
 - Data Bits: 0011
 - Parity Bit (LSB): 0
- **Analysis:**
 - Number of 1s: 2 (Even)
- **Result:** The parity checker validates the code as **Error-Free**.

XOR: $0 \oplus 0 \oplus 1 \oplus 1 \oplus 0$

$0 \oplus 0 \oplus 0 \oplus 0$

$0 \oplus 0 = 0$

$A \oplus B \oplus C \oplus D \oplus E = 0$
Error free

CASE STUDY 2 – ERROR DETECTED

- **Scenario:** A 4-bit message with an LSB parity bit.

- **Received Code:** 01011

- **Breakdown:**

- Data Bits: 0101

- Parity Bit (LSB): 1

- **Analysis:**

- Number of 1s: 3 (Odd)

- **Result:** The even parity checker identifies an **Error** because the total count of 1s is not even.

Good
01011 = 1 BCD + 1 P = 0
0 $\Rightarrow$ even

XOR
 $0 \oplus 1 \oplus 0 \oplus 1 \oplus 1$

$1 \oplus 1 \oplus 1$
 $0 \oplus 1 = 1$

HOW THE HARDWARE WORKS

- The Logic Gate: Even parity checkers are typically built using XOR (Exclusive-OR) gates.
- **Functionality:**
 - An XOR gate outputs a 1 if the number of inputs that are 1 is odd.
 - In an even parity system, if the XOR sum of all bits (including the parity bit) results in a 0, the data is valid.
 - If the result is 1, it indicates a parity error.